

Same Observations, Different Forecasts: Partition-Origin Sensitivity in Patch-Based Time-Series Forecasting

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Paper under double-blind review

Abstract

Patch-based time series forecasters assign the same observed history to a fixed lattice, yet the lattice origin is typically treated as an implementation detail. We vary only this origin while preserving every observed value, its temporal order, and the forecasting task. Under this observation-equivalent intervention, forecasts and errors vary with partition origin across all five examined benchmarks, with dataset-dependent magnitude. On the primary ETTh1 setting, checkpoints selected by validation exhibit a mean relative origin gap of $22.70 \pm 9.22\%$. Under a matched five-epoch control, the same sensitivity appears across three temporal mixers, and is attenuated by optimization state and pooled-head configuration. A token-axis spectral analysis of the latent token sequence characterizes how this sensitivity appears in learned representations and shows that it is reshaped rather than monotonically amplified through encoder depth. These findings highlight partition origin as an evaluation-relevant implicit coordinate choice in fixed-grid patch forecasting.

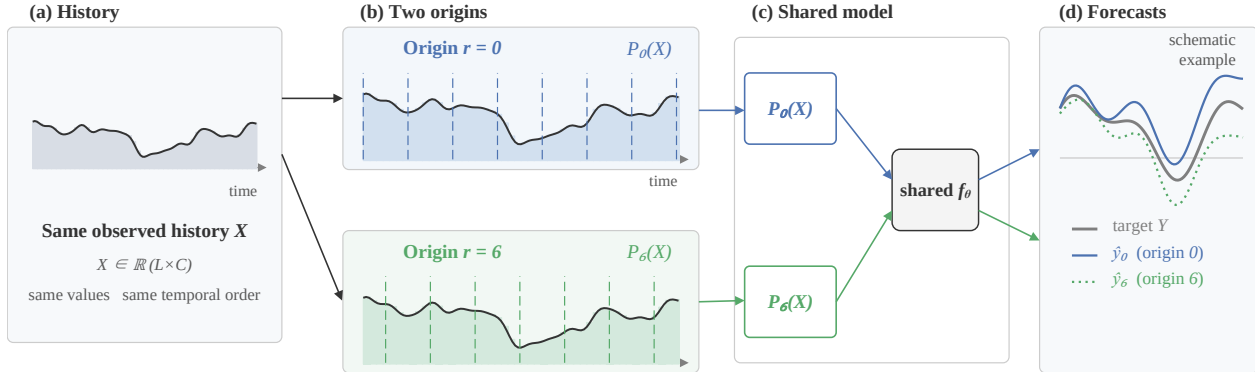


Figure 1: Schematic illustration of the partition-origin intervention. The same observed history is assigned to two patch lattices with different origins and processed by a shared forecaster. The illustrated forecast differences represent possible origin sensitivity, not measured model outputs. No timestamp shift, value perturbation, or data augmentation is applied.

1 Introduction

Time series forecasting underpins scheduling, risk management, and resource allocation in domains ranging from electricity load and price forecasting to weather, climate, traffic, and finance, where multi-step predictions of sensor and market streams inform downstream decisions. Much recent work has moved from classical statistical models toward learned representations of a fixed history. This makes representation construction part of the modeling problem: even when the observations, their order, and the forecasting target are held fixed, two observation-equivalent representations may expose different discrete coordinates to a forecaster.

The question studied here is therefore deliberately narrow. Given the same history X , model f_θ , and forecasting task, should forecasts agree when two patch maps $P_r(X)$ and $P_s(X)$ contain exactly the same real observations but assign them to different token and within-token coordinates? Such equality is not guaranteed because patch encoders generally assign observations to token indices and within-token positions that depend on lattice alignment.

Temporal patching groups neighboring observations into local segments, shortens the effective token sequence, and lets the forecaster operate on local temporal contexts rather than individual points. It is now used by attention, multilayer perceptron (MLP), and convolutional forecasters (Nie et al., 2023; Gong et al., 2023; Tang & Zhang, 2025). Channel-dependent and channel-independent models make different choices about interactions across variables (Nie et al., 2023; Liu et al., 2024); patching raises the separate question of how the history within each variable is coordinatized. PatchTST, for example, combines channel independence with temporal patching (Nie et al., 2023). Independently, strong results from simple channel-wise forecasters show that temporal representation choices can materially affect forecasting behavior (Zeng et al., 2023).

This representation strategy opens a broad design space. Models vary patch length, stride, overlap, hierarchy, semantic support, positional treatment, and downstream token processing (Huang et al., 2024; Chen et al., 2024; Peng et al., 2025; Zhang et al., 2024); other approaches adapt granularity, scale, boundaries, support, or token allocation to the signal (Ding et al., 2026; Ankireddy et al., 2026; Abeywickrama et al., 2025; Luo & Wang, 2024; Zucchi et al., 2026; Wu et al., 2025). These choices modify either the temporal partition or the processing applied to its tokens. They also reveal a more basic property of patching: each observation is assigned to a token and to a within-patch position, so the patch lattice acts as a discrete coordinate system induced by representation construction and presented to the model. Adaptive patching asks where and how to construct temporal patches. A separate question is whether, once a construction is fixed, its initial lattice alignment can itself affect forecasting outcomes. This distinction concerns the stability of a fixed representation rule, rather than selecting a more adaptive partition.

This sensitivity is not specific to time series. In computer vision, the discretization grid has long been known to act as a latent coordinate: padding and spatial-location studies show that grid boundaries provide implicit position cues (Islam et al., 2024; 2020), and work on shift equivariance and anti-aliasing shows that small changes in grid alignment can change network responses (Kayhan & van Gemert, 2020; Zhang, 2019). Recent vision-transformer studies make patch grid phase explicit through structured spatial offsets, inverse alignment, and aggregation, or through reflection-padded translations and phase marginalization (Ercan, 2026; Wang et al., 2026). The broader lesson is that a discretization grid can become a latent coordinate to which a model responds, even when that coordinate is not itself the task of interest. Unlike these vision and vision-language phase methods, which marginalize spatial patch phases for image representations, the present setting keeps the observed time series values and physical coordinates fixed while changing only their one-dimensional patch assignment.

Time series modeling has likewise examined stability under transformations that leave the forecasting task intact. Prior work studies invariance under transformations of a signal or its alignment (Vayer et al., 2022; Demirel & Holz, 2025), and Probingformer studies translation-related invariance at the patch level within a forecasting architecture (Zhou et al., 2026). Together, these studies motivate asking whether predictions depend on representation changes that preserve the forecasting task. Latent-space Fourier analysis has also been used to expose time-scale structure in learned representations for generative music (Wang & Huang, 2026). That setting applies frequency-domain control to audio-generation latents, whereas we use token-axis spectra only as a diagnostic of a fixed forecasting representation. The intervention considered here is narrower: it keeps the observed values, physical timestamps, temporal order, and forecasting target fixed while changing only their assignment to a one-dimensional patch lattice. Accordingly, the present contribution is an observation-equivalent audit of a representation coordinate, rather than a translation-invariant forecasting architecture. We do not claim that partition origin is the sole source of instability: we study it as a controlled representation perturbation and investigate how such perturbations are reflected in learned representations.

A further, often implicit degree of freedom is the partition origin r , the alignment between the observation window and the first patch of the lattice. The same observation can occupy token k and within-patch slot

i under one origin, but token k' and slot i' under another; boundary-mask geometry can change as well. Figure 1 schematically illustrates a possible forecast difference under this reassignment. We isolate this dependence with an intervention that changes only the lattice alignment rule while preserving observations and temporal order. Section 2 formalizes the protocol, and Appendix A gives the exact construction. The remainder of the paper asks whether forecasts actually change under this intervention, and if so, where that change becomes visible in the learned representation.

We additionally audit the official PatchTST implementation to examine whether the diagnostic recurs beyond the controlled research backbone. This public architecture check is intended to assess recurrence of the observation-equivalent diagnostic, not to validate POC or to create a standardized cross-model accuracy comparison; its raw-scale protocol and adapter are reported separately.

Our study makes three contributions: (1) an observation-equivalent partition-origin intervention that reveals sensitivity to an implicit coordinate choice in patch-based forecasting; (2) a representation-level analysis showing that changing the origin alters the latent token sequence and its spectral characteristics; and (3) a controlled evaluation across datasets, architectures, and an invariance-oriented intervention.

2 Problem Formulation

We call partition-origin sensitivity the dependence of a model’s forecast on the alignment of a fixed patch lattice under observation-equivalent representations of the same history.

2.1 Origin Intervention

Let $X \in \mathbb{R}^{L \times C}$ be a multivariate history, with forecasting target $Y \in \mathbb{R}^{H \times C}$. A fixed patch length p and origin $r \in \{0, \dots, p-1\}$ induce a partition $P_r(X)$, an ordered sequence of length- p tokens obtained by aligning a non-overlapping patch lattice at r and masking its boundary padding. The forecaster produces $\hat{Y}_r = f_\theta(P_r(X))$.

The intervention holds X , its temporal coordinates, its order, and the forecasting task fixed, and changes only r . Thus it is not a temporal translation of the signal. Changing r changes patch membership, within-patch coordinate assignment, and boundary-mask geometry, while leaving the real observations unchanged.

This distinction is important because a temporal translation changes the physical coordinate of the signal and can therefore change its relationship to the forecasting target, calendar effects, or data-generating process. Our intervention does none of these things: timestamps, values, ordering, and target remain fixed. It instead changes how an unchanged history is assigned to the discrete coordinates of a fixed patch representation. The resulting comparison is thus not a test of temporal shift invariance or a form of data augmentation; it is a controlled test of sensitivity to representation alignment.

2.2 Observation Equivalence

For every origin, each real observation appears exactly once in its original temporal order; none is duplicated or dropped. Boundary tokens retain their real observations and mask only padding slots. Invalid value slots are zeroed by the validity mask before projection, and the mask is concatenated as an additional input feature. Fully padded tokens are excluded from attention and zeroed after the encoder. Reconstruction, sentinel-invariance, and deterministic-evaluation checks confirm this behavior. Together, the masking protocol and checks rule out numerical padding values as the source of the observed sensitivity while preserving origin-dependent differences in boundary-mask geometry. Token construction and audit details are in Appendices A–B.

2.3 Metrics

For each origin r , let E_r be the test mean squared error (MSE) aggregated over test windows, channels, and the forecast horizon. We use two primary views. The prediction-level quantity

$$S_\theta(X) = \frac{1}{HC} \mathbb{E}_{r \neq s} \|\hat{Y}_r - \hat{Y}_s\|_F^2 \quad (1)$$

measures how much forecasts change for one history when only the representation origin changes, where the expectation is uniform over distinct ordered origin pairs (r, s) . We average it over test windows to obtain $\bar{S}_\theta = \mathbb{E}_X[S_\theta(X)]$; the pairwise identity in Appendix A relates it to the per-entry origin variance of the forecasts.

The task-level consequence is the relative origin gap

$$\Gamma_o = \frac{\max_r E_r - \min_r E_r}{\min_r E_r} \times 100\%. \quad (2)$$

Unlike $S_\theta(X)$, Γ_o measures variation in forecasting error and thus depends on the target as well as on the forecasts. Secondary diagnostics are defined in Appendix B. In particular, the interior gap Γ_i applies the same max–min comparison as Γ_o after excluding the special boundary origin $r=0$. The remaining diagnostics CV_o , Δ_o , E_0 , $E_\star$, $\bar{E}$, and E_{ens} supplement, rather than replace, the two primary views.

2.4 Coordinate Binding Hypothesis

Standard patch parameterizations do not enforce invariance across origins. A linearized patch encoder makes this coordinate binding explicit. For a single-channel history x_t , let $z_k^{(r)} \in \mathbb{R}^d$ be token k at origin r , $A_i \in \mathbb{R}^{d \times 1}$ its slot-specific projection, and $B_k \in \mathbb{R}^{H \times d}$ its token-specific contribution to the H -step forecast $\hat{y}_r \in \mathbb{R}^H$. Away from masked boundaries, write

$$z_k^{(r)} = \sum_{i=0}^{p-1} A_i x_{kp+i-r}, \quad \hat{y}_r = \sum_k B_k z_k^{(r)}. \quad (3)$$

An observation x_t contributes through an effective coefficient $B_k A_i$. Changing the origin can assign it to a different token and within-token slot, producing $B_{k'} A_{i'}$. Exact invariance would require agreement across every such reassignment, an equality not generally imposed by slot-specific patch projections and flattened token-index-specific heads. Pooling can weaken token-index dependence in B_k , whereas removing positional encoding removes only one coordinate cue. This is an illustrative parameterization and a mechanism hypothesis rather than an identified mechanism for the trained networks. It provides an intuitive explanation for why origin sensitivity can arise; the empirical controls below characterize how optimization and downstream head configuration modulate its magnitude.

3 Empirical Evidence

3.1 Main Phenomenon

Figure 2 summarizes the 15 dataset–replicate origin profiles. Formal gaps are computed from unnormalized per-origin errors; the window-level dispersion diagnostic in Figure 3 provides complementary evidence at the prediction level.

We first ask whether the observed sensitivity is confined to a particular dataset or appears across the examined forecasting benchmarks.

On primary ETTh1, the 15-epoch reference result selected by validation has $\Gamma_o = 22.70 \pm 9.22\%$ and $\Gamma_i = 16.29 \pm 3.40\%$. Origin zero is the largest-error origin in all three replicates, but the nonzero interior gap shows that the result is not solely attributable to the special $r=0$ layout.

Table 1 reports the three primary sensitivity summaries as three-replicate means and sample standard deviations across all five examined benchmarks. Variation appears on every benchmark but is heterogeneous: the ETT variants show stronger sensitivity, whereas Weather is weak relative to ETT.

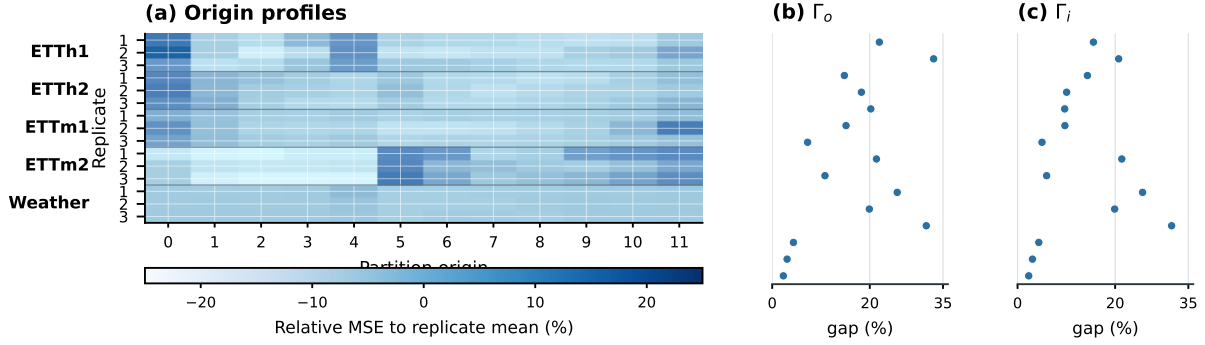


Figure 2: Origin profiles and aligned formal gaps across 15 dataset–replicate profiles. Lighter-to-darker blue denotes lower-to-higher normalized relative MSE in the origin-profile heatmap; the two right-hand columns report the full and interior gaps, Γ_o and Γ_i , on a common row ordering.

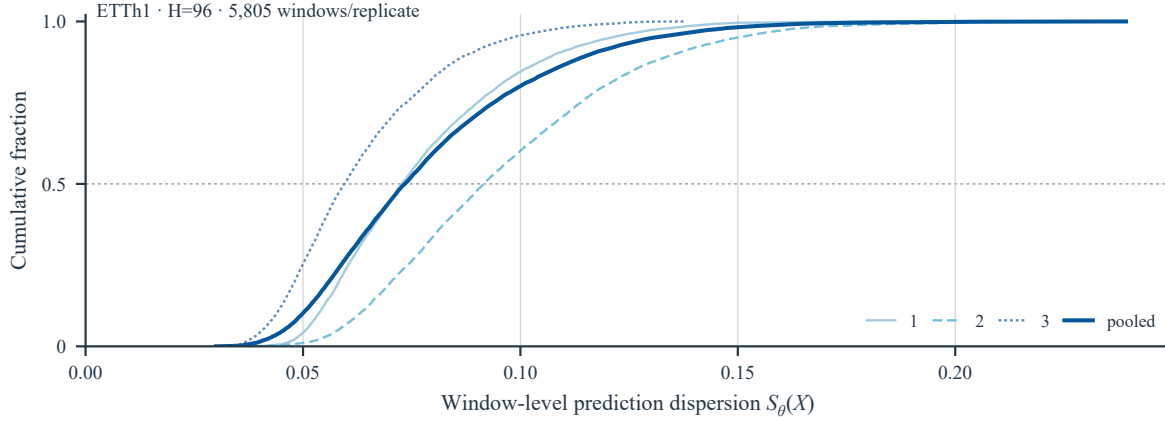


Figure 3: Window-level empirical cumulative distribution functions of $S_\theta(X)$ for three ETTh1 replicates and their pooled test windows.

Table 1: Core cross-dataset sensitivity summaries at $p=12$. Entries are means $\pm$ sample standard deviations over three independent replicates.

Dataset	Γ_o	Γ_i	$\bar{S}_\theta$
ETTh1	22.70 ± 9.22	16.29 ± 3.40	0.0754 ± 0.0210
ETTh2	17.30 ± 2.56	9.23 ± 0.22	0.0242 ± 0.0032
ETTh1	12.57 ± 7.34	10.20 ± 9.18	0.0315 ± 0.0130
ETTh2	25.12 ± 5.84	25.12 ± 5.84	0.0207 ± 0.0019
Weather	2.66 ± 1.04	2.66 ± 1.04	0.0044 ± 0.0004

3.2 Architectural and Training Factors

3.2.1 Across Architectures

We hold the ETTh1 representation, mask semantics, flattened head family, and complete primary split fixed while replacing the temporal mixer. This comparison examines whether the observed sensitivity extends beyond a specific architecture family, rather than claiming architecture-independent behavior. It uses a matched five-epoch budget and does not assess convergence. All nine replicates exhibit $\Gamma_o > 20\%$, indicating that the measured sensitivity persists under these matched mixer controls: the three-replicate mean Γ_o is 58.00% for the Transformer, 24.99% for the MLP, and 26.11% for the convolutional control (Appendix B.1).

A secondary audit indicates that origin choice can alter pairwise model margins in a controlled setting.

3.2.2 Optimization and Patch Geometry

Across saved checkpoints, the ETTh1 origin gap decreases overall but follows a non-monotonic trajectory (Figure 5). Thus optimization state affects the measured magnitude without supplying a convergence comparison. Changing patch length changes the measured Γ_o in both ETTh1 and Weather, but the ordering across $p=8, 12, 16$ is not stable across replicates. The largest ETTh1 value at $p=16$ is driven by a single boundary origin. Patch length modulates the measured magnitude without determining it monotonically in these diagnostics.

3.2.3 Downstream Configuration and Training Policy

Appendix B.1 reports the head diagnostic and training-origin policy comparisons: the masked flattened-head diagnostic gives $\Gamma_o = 8.68\%$ and the masked pooled head 1.21% , while boundary-only training gives $316.35 \pm 15.96\%$ and all-origin training $20.25 \pm 9.05\%$. The capped mask-head factorial changes both mask treatment and prediction-head configuration. Pooling yields a lower measured gap than the masked flattened-head diagnostic on both ETTh1 and Weather in this capped setting (Figure 5), providing evidence that prediction-head configuration can moderate the measured sensitivity. Because the comparison mixes one frozen diagnostic with multi-replicate summaries, we treat it as exploratory.

Training only at the boundary origin substantially worsens mean-origin error and sensitivity on ETTh1, showing that training-origin policy can materially change the measured sensitivity under all-origin evaluation in this setting.

Origin dependence is also observed under the stride-6 overlap lattice and at $H=192$ (Table 7; protocols in Appendix B.2).

3.3 Controls Against Common Alternative Explanations

The following controls do not identify a unique causal mechanism, but they test whether common implementation-level explanations are sufficient for the measured sensitivity. Reconstruction, padding-sentinel, and deterministic evaluation checks verify that every real observation is preserved and that changing numerical padding values does not account for the result; the origin-dependent boundary-mask geometry therefore remains part of the representation intervention rather than a numerical-padding artifact.

Explicit positional embeddings are not necessary in the primary ETTh1 setting: disabling their forward contribution yields $\Gamma_o = 24.42 \pm 6.90\%$, compared with $22.70 \pm 9.22\%$ in the primary condition (full run records are given in Appendix B.1). This control removes one coordinate cue, but not the token and within-patch assignments represented by B_k and A_i in Eq. 3; it does not support explicit positional embeddings as a sufficient explanation.

The observed sensitivity is also not restricted to one temporal mixer under the matched control: all nine Transformer, MLP, and convolutional replicates have $\Gamma_o > 20\%$. This supports persistence across these controlled architectures rather than an architecture-independent claim.

The nonzero interior gap on ETTh1 further shows that the primary result is not explained solely by the special $r=0$ boundary layout. The stride-6 overlap and $H=192$ extensions likewise retain origin dependence under targeted changes of patch geometry and forecast horizon. These are limited diagnostics, not exhaustive coverage of all patch constructions.

An external architecture audit provides additional evidence that the diagnostic can recur in a public patch-based implementation. With an explicit mask adapter around the official PatchTST source, the raw-scale protocol has $\Gamma_o = 326.86 \pm 40.53\%$ across three runs (Appendix B.2). This adapter-bound diagnostic is not a standardized MSE comparison with the primary controlled forecaster; it shows that the observation-equivalent audit can expose sensitivity in an official implementation under the stated adapter.

4 Representation-Level Evidence

4.1 Latent Token Sequence Spectral Analysis

To characterize where the observed discrepancy appears within the controlled Transformer, we analyze the three ETTh1 primary checkpoints on the same 5,805 test windows at origins 0 and 6. For each origin, we retain only its 42 fully observed patch tokens, take an FFT along the token axis of the latent channels, and compare spectral magnitudes per window. The FFT is taken along the token axis because that axis preserves the temporal ordering of the patch sequence, so its spectrum describes how latent variation is distributed over ordered patch positions rather than over embedding coordinates. Thus this diagnostic excludes partial and fully padded tokens; it does not treat the intervention as a temporal translation or test a Fourier phase-shift identity. We report the window-level Spearman association between latent magnitude difference and the MSE between the two forecasts.

More precisely, for test window i and origin r , let $Z_i^{(r)} \in \mathbb{R}^{K \times d}$ denote the final ordered token sequence after averaging model-variable copies when the encoder uses a variable-wise path, retaining only complete outer patches. Its row $z_{i,k}^{(r)}$ is the latent embedding of ordered token k , and j indexes an embedding coordinate, not a frequency axis. With $M = \lfloor K/2 \rfloor + 1$, we use the unitary real-valued token-axis DFT and compare magnitudes:

$$\widehat{Z}_{i,m,j}^{(r)} = \frac{1}{\sqrt{K}} \sum_{k=0}^{K-1} Z_{i,k,j}^{(r)} \exp(-i 2\pi m k / K), \quad m = 0, \dots, M-1, \quad (4)$$

$$D_{\text{spec},i} = \frac{1}{Md} \sum_{m=0}^{M-1} \sum_{j=1}^d \left| \left| \widehat{Z}_{i,m,j}^{(0)} \right| - \left| \widehat{Z}_{i,m,j}^{(6)} \right| \right|, \quad (5)$$

$$D_{\text{pred},i} = \frac{1}{HC} \left\| \widehat{Y}_i^{(0)} - \widehat{Y}_i^{(6)} \right\|_F^2, \quad \rho = \text{Spearman}(\{D_{\text{spec},i}\}_i, \{D_{\text{pred},i}\}_i). \quad (6)$$

The token order follows early-to-late patch positions, so m is an ordered-token frequency index. The comparison does not assume semantic alignment of token indices across origins; it measures changes in the ordered latent sequence induced by different lattice assignments. It is not a Fourier analysis along embedding coordinates or a physical-frequency decomposition of the original signal.

Table 2: Representation-level diagnostic summary for ETTh1, $p=12$, origins 0/6. Values are mean $\pm$ sample SD over three runs. DC is the $m=0$ coefficient; the other frequency entries average unsigned magnitude differences over their nonnegative token-DFT bins and hidden coordinates, so their absolute scales are not directly comparable across rows.

Analysis	Location or component	Median difference	ρ with forecast difference
Frequency	DC coefficient	0.722 ± 0.124	0.364 ± 0.072
	Non-DC low band	0.052 ± 0.001	0.363 ± 0.025
	Mid band	0.039 ± 0.001	0.221 ± 0.031
	High band	0.098 ± 0.006	0.545 ± 0.059
Layer	Post-position encoder input	0.109 ± 0.007	0.517 ± 0.040
	First encoder block	0.139 ± 0.005	0.508 ± 0.038
	Final normalized output	0.098 ± 0.005	0.529 ± 0.034

4.2 Frequency-wise Characterization

The large low-band difference is partly a DC component rather than evidence by itself of unstable long-term trend encoding: after separating DC, the non-DC low-band association with forecast difference is 0.363 ± 0.025 (Figure 4(a)). The high band has the strongest association (0.545 ± 0.059) in all three runs. Figure 4(a) also displays the individual seed values and their dispersion. These are representation-level associations in one controlled architecture, not a causal mediation analysis. A two-sided 1,000-permutation test of

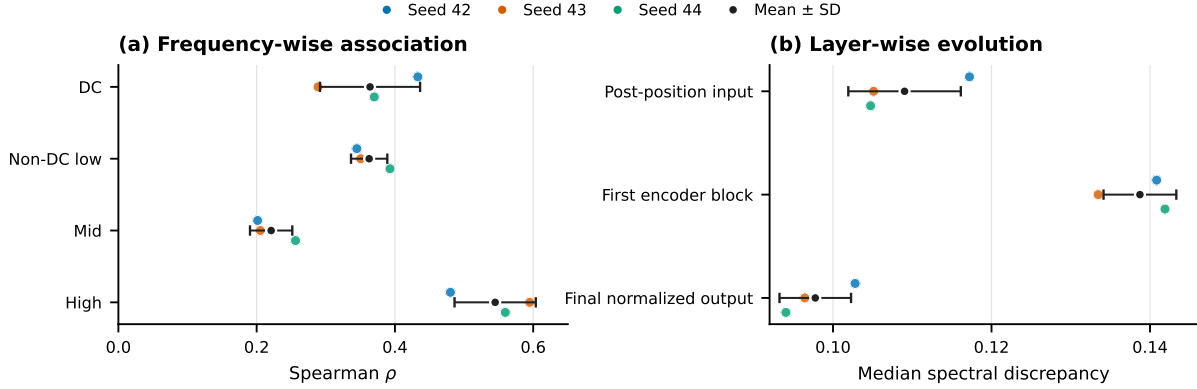


Figure 4: Frequency-wise and layer-wise summaries of the ETTh1 representation-level diagnostic across the three primary checkpoints. (a) Each colored point is the window-level Spearman association between a token-spectrum band difference and forecast disagreement for one seed; black circles and error bars are the mean and sample SD across seeds. (b) Colored points show the per-seed median token-spectrum discrepancy at the three extracted encoder locations; black circles and error bars show the mean and sample SD. These summaries describe association and representation differences in the controlled setting; they do not establish causal mediation.

random window pairing produces no equal-or-larger null correlation in any run (conservative $p = 0.000999$; Appendix B.5), making random pairing an unlikely explanation for the observed window-level association.

4.3 Layer-wise Evolution

Figure 4(b) shows that a discrepancy is already present at the post-position encoder input, peaks after the first block, and is smaller at the final output rather than growing monotonically. The discrepancy is therefore not simply accumulated with depth; it is reshaped by representation processing, so the final normalized output does not carry the largest layer-wise difference.

4.4 External Architecture Check

As an external architecture check, we repeat the final-encoder spectral diagnostic on the three frozen Protocol-Adapted PatchTST checkpoints used for the official-source audit (Section B.2), under the same ETTh1 raw-scale protocol and test windows. After discarding partial outer patches and the official embedder’s replication-padded token, each origin retains 42 complete tokens. The spectral-forecast Spearman association is positive in every run (0.189, 0.345, and 0.246; mean 0.260 ± 0.079), with each run’s bootstrap 95% interval excluding zero. The high-band association is likewise positive in all three runs (0.289, 0.336, and 0.313), although the DC component is larger in one run. This external-architecture check supports a protocol-bound recurrence of the diagnostic, not identical frequency behavior or numerical comparability with the standardized controlled Transformer. In particular, the correlation magnitude is not directly comparable because the implementations, masking adapters, and token pipelines differ.

5 Diagnostic Consistency Intervention

Partition-Origin Consistency (POC) is a diagnostic invariance-oriented intervention rather than a proposed method. It tests whether directly penalizing disagreement between observation-equivalent representations reduces held-out origin sensitivity, and whether that reduction aligns with forecasting accuracy.

5.1 Partition-Origin Consistency

For distinct origins r, s , POC evaluates the shared forecaster on $P_r(X)$ and $P_s(X)$. Writing $\hat{Y}_r = f_\theta(P_r(X))$ and $\hat{Y}_s = f_\theta(P_s(X))$, the training objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{cons}}, \quad (7)$$

where

$$\begin{aligned} \mathcal{L}_{\text{pred}} &= \frac{\|\hat{Y}_r - Y\|_F^2 + \|\hat{Y}_s - Y\|_F^2}{2HC}, \\ \mathcal{L}_{\text{cons}} &= \frac{\|\hat{Y}_r - \hat{Y}_s\|_F^2}{HC}. \end{aligned} \quad (8)$$

One distinct origin pair is sampled uniformly per mini-batch. The two views share parameters, data, and optimizer state. POC uses two forward passes during training, one ordinary forward pass at inference, and no added parameters. λ is selected by validation only; selection details are in Appendix A.4 and complete quantitative records in Appendix B.5.

We compare three training conditions. A is the single-view random-origin baseline. B uses a sampled pair of distinct origin views with the supervised prediction loss only. C uses the same two-view protocol as B and additionally applies the POC consistency term. B and C share initialization, origin-pair schedules, optimizer settings, and training budget; C versus B is the matched comparison and A is a single-view reference.

For a fixed predictor under deterministic evaluation, the sampled pairwise term in $\mathcal{L}_{\text{cons}}$ is an unbiased estimator of $S_\theta(X)$. The full identity and its proof are given in Appendix A.

5.2 Robustness and Accuracy

On ETTh1 at $p=12$, C reduces $\bar{S}_\theta$ by 57–60% relative to matched two-view control B in all three independent replicates, with Γ_o and Δ_o also reduced by 47–62% and average test error essentially unchanged or slightly improved. On ETTm2, the same ETTh1-selected weight lowers $\bar{S}_\theta$ by 30–66% without retuning, but error transfer is mixed: in one replicate C has larger Γ_o and Δ_o than B, and C has higher average MSE than A in all three replicates. In ETTm2 Replicate 3, for example, C has $\bar{E} = 0.287327$ versus 0.213736 for B, a 34.43% increase despite lower $\bar{S}_\theta$.

6 Conclusion

We identify a previously unreported sensitivity in patch-based forecasting: changing only the origin of a fixed patch lattice can change the forecast even when every observed value and its temporal order are preserved. The patch lattice assigns each observation to a token and within-patch coordinate, so partition origin is an evaluation-relevant implicit coordinate choice rather than a preprocessing detail. A token-axis spectral analysis indicates that origin-dependent differences are already observable in the latent token sequence and are reshaped rather than monotonically amplified through encoder depth, indicating that origin dependence is reflected in the representation rather than appearing only in a single downstream component. The consistency intervention adds that reduced disagreement and improved forecast accuracy need not coincide.

This matters for benchmarking and deployment. A comparison involving patch lengths, strides, tokenizers, or backbones may also be affected by an unreported lattice-alignment choice; protocols should therefore specify the origin policy used in training, validation, checkpoint selection, and inference, and audit a small set of observation-equivalent origins when those policies differ.

The present evidence covers $H=96$, non-overlapping patches, controlled backbones, and three replicates, with four of five core datasets from the ETT family; the matched-mixer study uses a five-epoch budget, and the pooled-head comparison does not separate capacity from optimization. The identified spectral association is not a complete causal decomposition of partition-origin sensitivity. Future work should study representation-level invariance to such coordinate choices.

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A Formalization and Experimental Details

This appendix records the partition construction, the pairwise-dispersion identity, and its implication for origin-dependent losses. For a fixed predictor under deterministic evaluation, the sampled pairwise term is an unbiased estimator of $S_\theta(X)$, and the following identity relates $S_\theta(X)$ to finite-origin variance.

A.1 Observation-Equivalent Partition Construction

For a history $X \in \mathbb{R}^{L \times C}$ and patch length p , the partition starting at origin $r \in \{0, \dots, p-1\}$ is $P_r(X)$. Two partitions $P_r(X)$, $P_s(X)$ are *observation-equivalent* (denoted $P_r(X) \sim_{\text{obs}} P_s(X)$) if they contain the same observed values, each exactly once, in the original time order, and with the same valid-observation coverage under the same masking rule. Under this equivalence, $P_r(X)$ and $P_s(X)$ differ in patch membership, within-patch coordinate assignment, and boundary-mask geometry.

For the non-overlapping construction used in the core study, let $K = \lceil (L + p - 1)/p \rceil$ and define, for $k = 0, \dots, K-1$ and $i = 0, \dots, p-1$,

$$\begin{aligned} t_{k,i}^{(r)} &= kp + i - r, & M_r(k, i) &= \mathbf{1}\{0 \leq t_{k,i}^{(r)} < L\}, \\ P_r(X)_{k,i,c} &= \begin{cases} X_{t_{k,i}^{(r)}, c}, & 0 \leq t_{k,i}^{(r)} < L, \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \quad (9)$$

The integer intervals $[kp - r, kp + p - 1 - r]$ partition the time indices, so every real observation is covered exactly once; the mask marks precisely the valid entries. The fixed token count is K for every origin. For $L=512$ and $p=12$, $K=44$, with r left-padding slots and $16 - r$ right-padding slots. The stride-6 extension in Appendix B.2 does not use this exact-once construction: overlapping tokens can contain the same observation more than once and are treated as a relaxed origin-controlled audit.

A.2 Origin Sensitivity Metrics

Recall $S_\theta(X)$ from Eq. 1. For the matrix-valued forecast $\hat{Y}_r = f_\theta(P_r(X))$, define the normalized total origin variance

$$\text{Var}_r[\hat{Y}_r] := \frac{1}{HC} \mathbb{E}_r \left[\|\hat{Y}_r - \mathbb{E}_r[\hat{Y}_r]\|_F^2 \right], \quad (10)$$

the mean squared distance from the origin-dependent forecast to its mean.

With $\bar{\hat{Y}} = \mathbb{E}_r[\hat{Y}_r]$,

$$S_\theta(X) = \frac{1}{HC} \mathbb{E}_{r \neq s} \|\hat{Y}_r - \hat{Y}_s\|_F^2 = \frac{2p}{p-1} \text{Var}_r[\hat{Y}_r]. \quad (11)$$

Proof. Let $\bar{\hat{Y}} = p^{-1} \sum_r \hat{Y}_r$. The standard pairwise-variance identity gives

$$\sum_{r \neq s} \|\hat{Y}_r - \hat{Y}_s\|_F^2 = 2p \sum_r \|\hat{Y}_r - \bar{\hat{Y}}\|_F^2.$$

Dividing by $p(p-1)HC$ yields Eq. 11.

Thus, for a fixed predictor under deterministic evaluation and a uniformly sampled distinct ordered pair (r, s) , $\|\hat{Y}_r - \hat{Y}_s\|_F^2 / (HC)$ is an unbiased estimator of $S_\theta(X)$.

To relate prediction disagreement to loss variation, *Remark.* Let $\mathbf{e}_r = \hat{Y}_r - Y$ and $\mathbf{e}_s = \hat{Y}_s - Y$ be per-origin forecast errors, and $\ell_r = \|\mathbf{e}_r\|_F^2$, $\ell_s = \|\mathbf{e}_s\|_F^2$ the corresponding squared Frobenius-norm errors over the $H \times C$ entries. Then

$$|\ell_r - \ell_s| \leq \|\hat{Y}_r - \hat{Y}_s\|_F (\|\mathbf{e}_r\|_F + \|\mathbf{e}_s\|_F). \quad (12)$$

This follows from the difference-of-squares identity, Cauchy–Schwarz, and the triangle inequality. Combined with the pairwise identity, it qualitatively links prediction consistency to differences between per-origin squared losses; it does not imply that the aggregate statistics Γ_o , CV_o , or Δ_o , or the mean per-origin error, decrease monotonically as the consistency penalty is strengthened.

Averaging across all origins removes dependence on a selected origin but requires p forward passes and does not make the underlying predictor origin-invariant. POC instead penalizes dispersion during training using two views per sample.

A.3 Model, Data, and Evaluation Configuration

The core study evaluates the ETT benchmarks ETTh1, ETTh2, ETTm1, and ETTm2 (Zhou et al., 2021), together with Weather (Max Planck Institute for Biogeochemistry, n.d.), using context length $L=512$, horizon $H=96$, patch length $p=12$, and non-overlapping stride 12, yielding 12 partition origins. All datasets use fixed chronological train/validation/test splits. Normalization statistics are fitted on training rows only and applied unchanged to validation and test data; metrics are computed in standardized space without inverse transformation. Exact splits and window counts appear in Table 3.

Table 3: Dataset splits and resulting window counts used in the core measurement.

Dataset	Train rows	Validation rows	Test rows	Train/Val/Test windows
ETTh1/ETTh2	[0, 8640)	[8640, 11520)	[11520, 17420)	8033 / 2785 / 5805
ETTm1/ETTm2	[0, 34560)	[34560, 46080)	[46080, 57600)	33953 / 11425 / 11425
Weather	[0, 36887)	[36887, 42157)	[42157, 52696)	36280 / 5175 / 10444

The standard ETT minute-resolution loader uses rows [0, 57600); the remaining source-file rows [57600, 69680) are not included in this protocol.

For each split, prediction targets are strictly contained within the corresponding split range, while historical context may extend to preceding observations as required by standard forecasting protocols.

The primary quantities in Section 2.3 are the aggregate prediction dispersion $\bar{S}_\theta$ and the relative origin gap Γ_o . For completeness, the retained secondary error diagnostics include the coefficient of variation over origins, denoted CV_o , and the absolute origin spread Δ_o :

$$\text{CV}_o = \frac{\text{std}_r(E_r)}{\text{mean}_r(E_r)}, \quad (13)$$

$$\Delta_o = \max_r E_r - \min_r E_r. \quad (14)$$

Here std_r is the population standard deviation over the finite set of origins; across independent runs we report a sample standard deviation. Since origin 0 has maximized trailing padding, we also retain the interior diagnostic

$$\Gamma_i = \frac{\max_{r \geq 1} E_r - \min_{r \geq 1} E_r}{\min_{r \geq 1} E_r} \times 100\%, \quad (15)$$

which is secondary and does not replace Γ_o . This diagnostic removes only the special $r=0$ layout; it does not remove all origin-dependent boundary-mask geometry. The inference summaries are origin-zero error E_0 , and error at the origin selected by validation $E_\star$ with $r^\star = \arg \min_r E_r^{\text{val}}$, mean-origin error $\bar{E} = p^{-1} \sum_r E_r$, and prediction-ensemble error E_{ens} ; the latter averages predictions before computing MSE and requires p forward passes.

We use a controlled channel-independent masked-patch Transformer as a research backbone rather than a reimplementaion of a named forecasting architecture. Patch values are concatenated with binary validity masks before projection, and fully padded tokens follow the masking protocol in Section 2.2. Training samples one batch-shared origin uniformly, and checkpoints are selected by validation MSE averaged over all 12 origins. Concretely, the backbone uses a linear tokenizer ($2p \rightarrow 64$) over patch values concatenated with the binary validity mask, producing 44 tokens at $p=12$. The encoder has two layers, four heads, FFN width 256, post-norm activations, and ReLU, and the flattened head maps $44 \times 64 \rightarrow 96$ per channel. Optimization

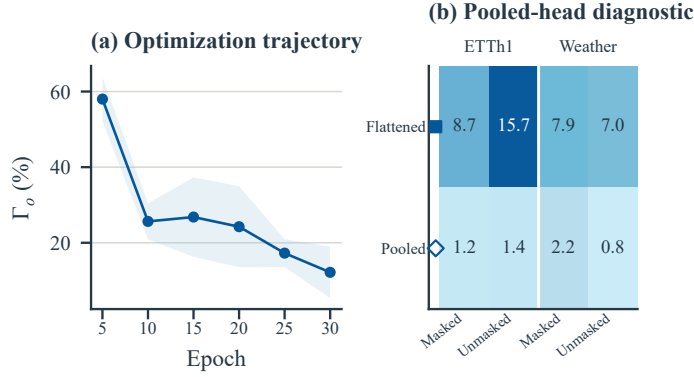


Figure 5: Optimization and pooled-head moderator diagnostics. The optimization line and shaded region show the mean $\pm$ sample SD over three replicates; Γ_o decreases from $58.00 \pm 5.84\%$ at epoch 5 to $12.19 \pm 6.80\%$ at epoch 30. In the matrix, masked flattened cells are single frozen contrasts and the remaining cells are three-run means; it is an exploratory diagnostic, not a matched backbone comparison. Cell values are Γ_o percentages.

uses AdamW with learning rate 10^{-4} , weight decay 10^{-4} , batch size 32, and 15 epochs, with dropout 0.1, gradient clipping 1.0, and no learning-rate scheduler. Learnable zero-initialized 44×64 token positions are added after projection, and checkpoints are selected by the lowest mean validation MSE over the 12 origins, breaking exact ties toward the later epoch.

A.4 Consistency Objective and Selection

On predesignated Replicate 1, $\lambda \in \{0.01, 0.03, 0.1, 0.3, 1.0\}$ is selected by validation only. A candidate is eligible only when its validation mean MSE satisfies $\bar{E}_C^{\text{val}} \leq 1.01 \bar{E}_B^{\text{val}}$; among eligible candidates, we select the smallest validation $\bar{S}_\theta$ (breaking an exact 10^{-12} tie toward the smaller λ). The predesignated B value is 0.748989, giving an eligibility limit of 0.756479; all five candidates pass, and $\lambda=1.0$ minimizes validation $\bar{S}_\theta$. No test metric enters selection, and B and C use matched initialization and origin-pair schedules.

Table 4: Validation-only selection of the consistency weight on the ETTh1/ $p=12$ development replicate. Eligibility requires validation mean MSE no larger than 1.01 times the B-condition value.

λ	Val. $\bar{E}$	Val. $\bar{S}_\theta$
0.01	0.746071	0.063589
0.03	0.745142	0.061730
0.1	0.742391	0.056330
0.3	0.734339	0.043623
1.0	0.735661	0.025627

B Additional Experiments and Quantitative Results

B.1 Architecture, Optimization, and Training Controls

Table 5 gives the per-replicate metrics for the full-split ETTh1 control of Section 3.2. This study follows the ETTh1 data, origin, and evaluation protocol of Appendix A.3, with a matched five-epoch training budget. The three research backbones use the same 64-dimensional tokenizer and flattened 44-token head, and are parameter-matched to within 0.001M: the Transformer uses two layers, four heads, and FFN width 256; the MLP uses two residual blocks with 44-slot mixing; and the Conv backbone uses two residual blocks with a $k=3$ temporal convolution, giving 0.375M, 0.345M, and 0.375M parameters respectively. These are research controls rather than implementations of named public forecasting architectures. The matched five-epoch budget is reported descriptively; several replicates select the final permitted epoch, so this is not

a convergence study. The saved curves show continued improvement at the five-epoch budget boundary: validation MSE decreases through epoch 5 for the Transformer and Conv runs, while two MLP runs select epoch 4.

Optimization attenuates the measured gap overall, although the trajectory is non-monotonic. The pooled-head configuration also shows lower sensitivity in this controlled diagnostic (Figure 5).

Table 5: Core matched-backbone evidence under the full-split five-epoch protocol.

Backbone	Rep.	$\bar{E}$	Γ_o
Transformer	1	0.671718	57.720%
	2	0.691658	63.976%
	3	0.696986	52.310%
MLP	1	0.598638	24.333%
	2	0.617356	26.899%
	3	0.594813	23.749%
Conv	1	0.613500	25.816%
	2	0.602558	26.263%
	3	0.597210	26.256%

The no-positional-encoding control retains the learnable 44×64 positional parameter and the matched initialization, but disables its addition to projected patch tokens in the forward pass. It otherwise uses the same split, random-origin training, optimizer, 15-epoch budget, and mean-over-origins validation selection as the primary protocol. The three runs select epochs 14, 12, and 15, with $\bar{E}$ of 0.500379, 0.527218, and 0.502761 and Γ_o of 25.665220%, 30.601097%, and 16.980709%; their aggregate is $\bar{E} = 0.510119 \pm 0.014856$ and $\Gamma_o = 24.415675 \pm 6.895633\%$.

The Figure 5 matrix is a separate capped diagnostic on ETTh1 and Weather, not a rerun of the 15-epoch canonical study or the full-split five-epoch mixer study. Its four columns correspond to A, masked flattened head; B, unmasked flattened head; C, masked pooled head; and D, unmasked pooled head. A is a frozen single-diagnostic contrast, whereas B–D are arithmetic means of the three run-level records for each dataset; the matrix is therefore not a replicate-level table or a directly matched estimate of A against B–D. The frozen artifact contains complete three-run records for B–D on both datasets, whereas A remains a single-diagnostic contrast. All four conditions use $L=512$, $H=96$, $p=12$, stride 12, a 44-slot tokenizer, batch size 32, 1,024/16/16 train/validation/test windows, and the stated five-epoch cap. B–D share the tokenizer, optimizer, replicate set, and valid-observation construction; mask treatment is the one indicated by each column, and each is evaluated from its checkpoint selected by validation within that cap. Capacity and optimization are not separately manipulated.

B.2 Additional Origin-Controlled Evaluations

These supplementary controlled audits extend the primary cross-dataset measurement. Unless stated otherwise, they inherit the canonical protocol of Appendix A.3; only deviations are reported below. All reported results passed independent recomputation and padding-sentinel checks.

This audit substitutes the official PatchTST implementation for the controlled backbone, trained with its native Adam configuration for 10 epochs. The observation-equivalent intervention produces partially padded windows, whereas the official implementation assumes complete ones; the audit therefore wraps it in a mask-aware adapter that computes per-window statistics from valid observations only.¹ For a channel with validity

¹The adapter is implemented in `runner.py`; the forecaster source is the official `models/PatchTST.py`.

mask m , the adapter normalizes

$$\begin{aligned} x_{\text{safe}} &= m \odot x, & n &= \sum_t m_t, \\ \mu &= \frac{1}{n} \sum_t x_{\text{safe},t}, & \sigma^2 &= \frac{1}{n} \sum_t m_t (x_{\text{safe},t} - \mu)^2 + 10^{-5}, \\ \tilde{x}_t &= \frac{m_t (x_{\text{safe},t} - \mu)}{\sigma}, \end{aligned} \quad (16)$$

so that padded positions are zero after normalization. The normalized window is then passed to the official patch embedding, encoder, and prediction head, and the output is mapped back to the raw data scale as

$$\hat{Y}_{\text{raw}} = \hat{Y}_{\text{norm}} \odot \sigma + \mu \quad (17)$$

before losses and metrics are computed. The inverse transform is thus applied exactly once, and the official forecast-path normalization is not applied a second time. Native inner patch padding is retained and the mask is not appended as an additional input. Because the reported values are raw-scale, they must not be compared numerically with the standardized controlled-Transformer MSE values.

Across the three runs, the raw-scale protocol gives $\bar{E} = 29.24 \pm 2.14$, $\Gamma_o = 326.86 \pm 40.53\%$, $\Gamma_i = 283.36 \pm 30.73\%$, $\Delta_o = 33.84 \pm 3.93$, and $\text{CV}_o = 0.418 \pm 0.018$.

This applies the audit to an official PatchTST source with an explicit protocol adapter; the reported values are interpreted on that adapted basis. The large relative gap is an audit diagnostic for this raw-scale protocol, not a standardized comparison with the primary controlled forecaster.

The main-text spectral diagnostic reuses these exact ETTh1 checkpoints at origins 0/6. It retains only the 42 complete outer tokens per origin and excludes the official embedder’s extra replication-padded token; its per-window records are therefore a representation diagnostic of this adapted protocol rather than a modification of the official source or its original audit metrics.

This audit tests whether the training-origin policy moderates origin sensitivity, under the data, preprocessing, origin, evaluation, and controlled-Transformer settings of Appendix A.3. The random-origin condition retains the canonical one-origin-per-batch sampling rule. Boundary-only training always uses $r=0$; random-origin training samples one batch-shared origin uniformly; and all-origin training evaluates all 12 origins for each batch, averages their losses, and then takes one optimizer step. The all-origin policy uses more forward passes per optimizer step, so these policies are not treated as compute-matched. Training at a fixed boundary origin worsens origin sensitivity, while random-origin and all-origin training are substantially less sensitive under this setting.

Table 6: Training-origin policy comparison. Entries are means $\pm$ sample standard deviations over three runs.

Strategy	$\bar{E}$	$\Gamma_o(\%)$	$\Gamma_i(\%)$	Δ_o	CV_o
Boundary-only	1.247804 ± 0.015822	316.349775 ± 15.963303	267.458415 ± 8.475230	1.400811 ± 0.037559	0.408476 ± 0.004946
Random-origin	0.504599 ± 0.014760	24.700318 ± 9.212623	14.898377 ± 4.138212	0.117916 ± 0.044519	0.064652 ± 0.021264
All-origin	0.498532 ± 0.016074	20.249528 ± 9.053775	13.067810 ± 6.238475	0.096813 ± 0.044753	0.055102 ± 0.024773

This comparison identifies training-origin policy as a moderator under all-origin evaluation in the stated setting.

This audit extends the measurement to an overlapping lattice. Relative to Appendix A.3, only the stride changes: stride 12 has origins 0–11 and 44 patches in the padded sequence, whereas stride 6 has origins 0–5 and 86 patches. The stride-6 audit uses a separately defined six-origin lattice; Table 7 reports the result.

This audit extends the measurement to a longer forecast horizon. Relative to Appendix A.3, only the horizon changes, from $H=96$ to $H=192$, which yields the horizon-specific split 7,937/2,689/5,709 windows. Other data processing, origin, and optimizer settings are inherited from the canonical protocol. Table 7 reports the resulting sensitivity alongside the overlapping-lattice extension.

Table 7: Origin sensitivity under two lattice extensions: a stride-6 overlapping lattice and a longer forecast horizon ($H=192$), each over three runs. Entries are means $\pm$ sample standard deviations.

Metric	Overlap (stride 6)	Longer horizon ($H=192$)
$\bar{E}$	0.525810 ± 0.023775	0.570299 ± 0.021573
Γ_o	$27.393200 \pm 15.299796\%$	$23.637704 \pm 5.120640\%$
Γ_i	$23.589134 \pm 17.747020\%$	$17.164232 \pm 3.290173\%$
Δ_o	0.129960 ± 0.071283	0.125384 ± 0.029311
CV_o	0.087099 ± 0.040541	0.060399 ± 0.008478

B.3 Consistency Regularization Results

Table 8 reports the ETTh1 condition-level results, including absolute and reduction values.

Table 8: ETTh1 condition-level results for the consistency intervention. The C/B columns give the relative change of condition C against matched control B.

cond.	Replicate	$\bar{E}$	$\bar{S}_\theta$	Γ_o	CV_o	Δ_o	$\bar{S}_\theta$ C/B	Γ_o C/B	Δ_o C/B	E_{ens}
A	1	0.494582	0.067678	21.38%	0.0600	0.100392	—	—	—	0.463563
B	1	0.495702	0.064642	20.24%	0.0602	0.095091	—	—	—	—
C	1	0.493625	0.026339	10.53%	0.0303	0.050582	-59.25%	-47.99%	-46.81%	—
A	2	0.528315	0.099207	32.50%	0.0821	0.157043	—	—	—	0.482845
B	2	0.529198	0.098973	39.63%	0.1010	0.196825	—	—	—	—
C	2	0.516281	0.039389	15.07%	0.0477	0.075153	-60.20%	-61.98%	-61.82%	—
A	3	0.495647	0.059302	14.21%	0.0420	0.066979	—	—	—	0.468467
B	3	0.493262	0.049484	12.28%	0.0359	0.058302	—	—	—	—
C	3	0.481394	0.021232	5.24%	0.0141	0.024671	-57.09%	-57.36%	-57.68%	—

Table 9 reports the ETTm2 transfer results, including absolute values and the relative change of condition C against matched control B.

Table 9: ETTm2 transfer results for the consistency intervention. The C/B columns give the relative change of condition C against matched control B.

cond.	Replicate	$\bar{E}$	$\bar{S}_\theta$	Γ_o	CV_o	Δ_o	$\bar{S}_\theta$ C/B	Γ_o C/B	Δ_o C/B
A	1	0.205168	0.021247	25.05%	0.0839	0.045571	—	—	—
B	1	0.303756	0.022616	29.84%	0.1041	0.078109	—	—	—
C	1	0.266032	0.007695	14.75%	0.0471	0.036286	-65.98%	-50.57%	-53.54%
A	2	0.199453	0.018643	19.33%	0.0538	0.036031	—	—	—
B	2	0.252393	0.016077	24.06%	0.0619	0.054849	—	—	—
C	2	0.234219	0.006611	11.34%	0.0342	0.025083	-58.88%	-52.88%	-54.27%
A	3	0.217110	0.022335	31.00%	0.0862	0.058784	—	—	—
B	3	0.213736	0.025060	35.64%	0.1045	0.065647	—	—	—
C	3	0.287327	0.017439	45.13%	0.1238	0.107310	-30.41%	+26.65%	+63.46%

B.4 Qualitative Paired PCA Visualization of Origin-Induced Latent Displacements

Figure 6 provides a qualitative companion to the token-spectrum summaries. It uses only the frozen ETTh1 primary seed-42 checkpoint and pools the complete tokens of the same 72 test windows under origins 0 and 6. The windows are sampled deterministically from low, middle, and high forecast-disagreement terciles (24 each). Each panel fits a separate PCA to the concatenated 144 pooled representations, so its axes are panel-specific. The paired lines visualize projected displacement only: their lengths are not full-dimensional latent distances, and the figure does not establish causality. In particular, the construction does not assume semantic alignment of token indices across origins.

B.5 Detailed Quantitative Results and Reproducibility

Detailed cross-dataset, per-origin, and patch-length results are included in this appendix.

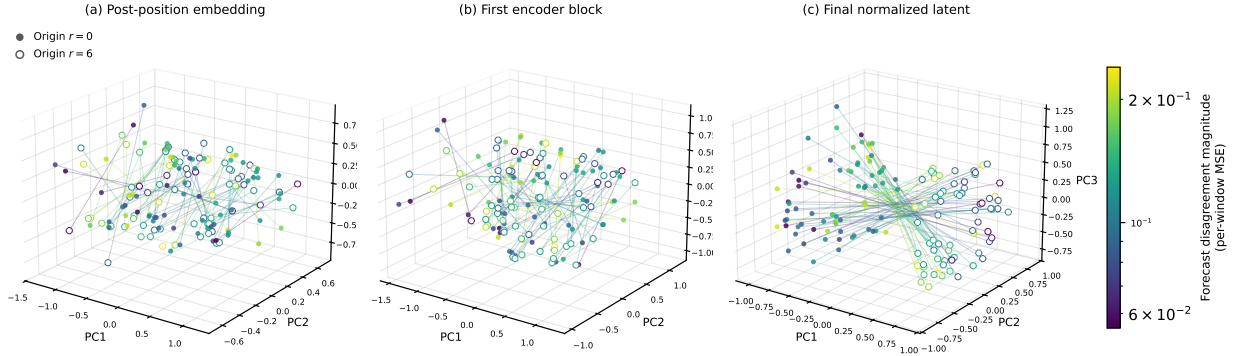


Figure 6: Qualitative paired PCA visualization of origin-induced latent displacements. Each panel shows a 3D PCA projection of pooled window-level latent representations for the same selected ETTh1 test windows under partition origins $r=0$ and $r=6$, using the frozen seed-42 primary checkpoint. Each line connects the two projected representations of the same observation window. Colors encode the paired forecast-disagreement magnitude (per-window MSE). Because each panel is a separate PCA projection, projected distances visualize qualitative displacement only and are not directly comparable across panels or equal to full-dimensional latent distances.

To test the significance of the latent spectral association, we keep the 5,805 per-window forecast discrepancies fixed and randomly permute the paired token-spectrum discrepancies across windows 1,000 times. We recompute the same Spearman statistic for every permutation and report a conservative two-sided Monte Carlo p value, $(b+1)/(1000+1)$, where b is the number of null statistics whose absolute value is at least the observed absolute correlation. This tests random pairing, not causal mediation.

Across the three runs, the observed ρ is 0.493946, 0.532947, and 0.560852, against permutation means of -0.000220 ± 0.013016 , 0.000041 ± 0.012563 , and -0.000115 ± 0.012650 ; every two-sided p is 0.000999.

For pairwise model margins, define $d_r = E_A(r) - E_B(r)$ from the frozen per-origin errors of two matched backbones. A strict crossing is recorded only when $\min_r d_r < 0 < \max_r d_r$; an exact zero tie alone is not a reversal. Every crossing found is in Weather Replicate 1: Transformer-MLP crosses at origin 7 (d_r ranges -0.03302 to 0.00134), Transformer-Conv at origins 8–11 (-0.02069 to 0.00689), and MLP-Conv at origins 1–4, 6, and 7 (-0.00861 to 0.01845). No crossing is observed in ETTh1 Replicates 1–3 or Weather Replicates 2–3.

The numerical records permit checking the reported aggregates and metric relationships at the displayed precision. Protocol-specific reconstruction, sentinel-invariance, deterministic-evaluation, and artifact checks are reported with their experiments. Full verification of data processing, training, and sentinel procedures would require the code and artifacts and is outside the scope of the present evidence.

Table 10 preserves the full cross-dataset inference and dispersion records for all 15 primary dataset–replicate records.

Table 11 reports the complete ETTh1 optimization trajectory as the mean $\pm$ sample standard deviation over three independent replicates; Figure 5 is the analytical summary.

Table 10: Full cross-dataset records for all 15 primary dataset–replicate entries. Top: inference summaries. Bottom: dispersion summaries.

<i>Core inference records.</i>							
Dataset	Replicate	r^*	E_0	E_*	$\bar{E}$	E_{ens}	$\bar{S}_\theta$
ETTh1	1	6	0.569854	0.480050	0.494582	0.463563	0.067678
ETTh1	2	6	0.640240	0.493491	0.528315	0.482845	0.099207
ETTh1	3	1	0.538284	0.471305	0.495647	0.468467	0.059302
ETTh2	1	8	0.305999	0.259954	0.274247	0.261520	0.027766
ETTh2	2	7	0.306528	0.256233	0.270232	0.260312	0.021643
ETTh2	3	7	0.293679	0.256367	0.266448	0.255766	0.023307
ETTm1	1	5	0.351157	0.329183	0.334726	0.323829	0.023775
ETTm1	2	6	0.371770	0.319341	0.339151	0.317818	0.046544
ETTm1	3	7	0.345944	0.314875	0.322839	0.311750	0.024194
ETTm2	1	2	0.191459	0.181952	0.205168	0.195430	0.021247
ETTm2	2	7	0.197801	0.204016	0.199453	0.190909	0.018643
ETTm2	3	7	0.214728	0.217987	0.217110	0.206873	0.022335
Weather	1	10	0.149571	0.149651	0.149692	0.147542	0.004692
Weather	2	5	0.150821	0.149475	0.150915	0.149085	0.003993
Weather	3	7	0.151174	0.150546	0.151079	0.149006	0.004525
<i>Core dispersion records.</i>							
Dataset	Replicate		Γ_o	CV_o	Δ_o	Γ_i	ident. err.
ETTh1	1		21.384539	0.060006	0.100392	14.975795	1.6×10^{-10}
ETTh1	2		32.500937	0.082108	0.157043	20.142130	8.7×10^{-10}
ETTh1	3		14.211458	0.042029	0.066979	13.739461	2.0×10^{-10}
ETTh2	1		17.712599	0.044050	0.046045	9.485186	2.4×10^{-10}
ETTh2	2		19.628638	0.046547	0.050295	9.082970	1.2×10^{-10}
ETTh2	3		14.554123	0.041395	0.037312	9.129265	9.8×10^{-11}
ETTm1	1		6.675168	0.019387	0.021974	4.432710	1.1×10^{-10}
ETTm1	2		20.787054	0.059335	0.066382	20.787054	5.4×10^{-11}
ETTm1	3		10.237248	0.026589	0.032126	5.382824	8.1×10^{-12}
ETTm2	1		25.045514	0.083859	0.045571	25.045514	1.4×10^{-10}
ETTm2	2		19.326828	0.053835	0.036031	19.326828	3.3×10^{-11}
ETTm2	3		30.999016	0.086162	0.058784	30.999016	2.2×10^{-10}
Weather	1		3.777446	0.009141	0.005598	3.777446	5.9×10^{-12}
Weather	2		2.478596	0.006775	0.003702	2.478596	2.6×10^{-11}
Weather	3		1.713865	0.004381	0.002568	1.713865	6.7×10^{-12}

Table 11: ETTh1 optimization trajectory. Entries are means $\pm$ sample standard deviations over three replicates.

Epoch	$\bar{E}$	Γ_o	Γ_i	$\bar{S}_\theta$
5	0.686787 $\pm$ 0.013320	58.002 $\pm$ 5.838%	53.207 $\pm$ 9.616%	0.082535 $\pm$ 0.005360
10	0.532454 $\pm$ 0.022974	25.651 $\pm$ 4.771%	18.996 $\pm$ 7.174%	0.044652 $\pm$ 0.011792
15	0.511572 $\pm$ 0.011890	26.805 $\pm$ 10.511%	16.707 $\pm$ 4.284%	0.036197 $\pm$ 0.007556
20	0.514882 $\pm$ 0.002471	24.252 $\pm$ 10.706%	10.528 $\pm$ 1.589%	0.032265 $\pm$ 0.008142
25	0.515072 $\pm$ 0.002251	17.262 $\pm$ 3.687%	10.815 $\pm$ 8.178%	0.032116 $\pm$ 0.004721
30	0.516227 $\pm$ 0.006846	12.187 $\pm$ 6.796%	12.165 $\pm$ 6.820%	0.029609 $\pm$ 0.005437

Table 12: Primary ETTh1 per-origin test MSE for the three replicates.

Replicate \ origin	0	1	2	3	4	5	6	7	8	9	10	11
1	0.5699	0.4879	0.4748	0.5092	0.5398	0.4852	0.4800	0.4785	0.4734	0.4695	0.4706	0.4962
2	0.6402	0.5215	0.4832	0.5064	0.5805	0.4992	0.4935	0.5013	0.5008	0.5215	0.5288	0.5626
3	0.5383	0.4713	0.4767	0.5018	0.5361	0.4980	0.4963	0.4896	0.4814	0.4797	0.4796	0.4989

Table 13: Complete patch-length sensitivity values for ETTh1 and Weather.

Dataset	p	Replicate	sel. epoch	$\bar{E}$	Γ_o	CV_o	Δ_o	Γ_i
ETTh1	8	1	12	0.504245	13.30%	0.03649	0.064506	4.79%
ETTh1	8	2	12	0.506219	15.46%	0.04700	0.075636	6.03%
ETTh1	8	3	14	0.529534	20.51%	0.06577	0.102419	14.78%
ETTh1	12	1	14	0.494582	21.38%	0.06001	0.100392	14.98%
ETTh1	12	2	11	0.528315	32.50%	0.08211	0.157043	20.14%
ETTh1	12	3	14	0.495647	14.21%	0.04203	0.066979	13.74%
ETTh1	16	1	14	0.508278	37.80%	0.08386	0.183037	9.24%
ETTh1	16	2	14	0.586954	118.16%	0.24045	0.610103	14.24%
ETTh1	16	3	13	0.540747	27.63%	0.05946	0.140124	10.23%
Weather	8	1	7	0.153532	1.76%	0.00645	0.002681	1.76%
Weather	8	2	10	0.154419	2.18%	0.00613	0.003345	1.32%
Weather	8	3	10	0.150530	2.14%	0.00766	0.003189	2.14%
Weather	12	1	12	0.149692	3.78%	0.00914	0.005598	3.78%
Weather	12	2	13	0.150915	2.48%	0.00678	0.003702	2.48%
Weather	12	3	11	0.151079	1.71%	0.00438	0.002568	1.71%
Weather	16	1	15	0.152961	5.80%	0.01137	0.008684	3.31%
Weather	16	2	9	0.151383	10.40%	0.02381	0.015539	2.42%
Weather	16	3	15	0.151240	5.15%	0.01193	0.007696	2.56%